Bachelor of Science in Business Administration Undergraduate FALL Term

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TUNIS BUSINESS SCHOOL
UNIVERSITY OF TUNIS

IT360: Information Security

IT 360 Project

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High-Level Design (HLD)

The proposed solution is a secure, end-to-end image steganography system designed to protect the existence of sensitive communication. It integrates Encryption with Steganography to provide two layers of protection.

Core Components

The system consists of four primary functional components:

- **Cover Medium:** The original digital image (JPEG or PNG) used to carry the hidden data.
- **Preprocessing Engine:** Converts the secret message into binary form and applies encryption.
- **Stego-Engine (Encoder/Decoder):** The central logic that embeds data into pixel values using the LSB method.
- **Security Controller:** Manages the Stego Key (password), which is used to encrypt and decrypt the hidden message.

Data Exchange Flow

Phase	Exchanged Data / Action	Description
Sender Input	Secret Message + Cover Image	The user provides the message to hide and the carrier image.
Encoding	Message → XOR Encrypt → Binary → Stego Image	The message is encrypted with XOR, converted to binary, and embedded into the LSB of each pixel's RGB channels.
Transmissio n	Stego Image (PNG)	An image that looks visually unchanged is sent to the receiver. Saved as PNG to preserve lossless pixel data.
Decoding	Stego Image + Key → Message	The receiver uses the same password to extract and decrypt the original message.

Proposed Tools & Technology Stack

Development Tools

During the research phase, several existing tools were evaluated including Steghide (transform domain) and StegoSuite (spatial domain). After evaluation, a custom Python implementation was selected to allow full control over the embedding and encryption logic.

The final implementation uses:

- **Python 3** — core programming language
- **Pillow (PIL)** — image processing library for reading and writing pixel data
- **LSB (Least Significant Bit)** — spatial domain embedding method, selected for its simplicity, efficiency, and high embedding capacity
- **XOR Cipher** — applied as the encryption layer before embedding. Note: XOR with a repeating key is used here for demonstration purposes. In a production system, AES encryption would be recommended for stronger security.

Evaluation Metrics

- **PSNR (Peak Signal-to-Noise Ratio):** Computed in the implementation to measure the visual difference between the original and stego image. A result above 40 dB confirms the changes are imperceptible to the human eye.
- **SSIM (Structural Similarity Index):** Identified as a recommended metric for future work to further validate imperceptibility.

Development Phases

Phase 1: Research & Requirements (Analysis)

- Reviewed existing steganography techniques across spatial domain (LSB, PVD), transform domain (DCT, DWT), and AI-based methods (CNNs, GANs).
- Evaluated the four core trade-offs: Capacity, Robustness, Security, and Imperceptibility.
- **Decision:** LSB spatial domain embedding was selected for the proof-of-concept due to its simplicity, low computational cost, and suitability for a controlled environment.

Phase 2: Core Implementation (Development)

- **Encryption Layer:** XOR cipher applied to the message before embedding, ensuring that even if the hidden data is detected, it remains unreadable without the password.
- **Embedding Algorithm:** LSB injection implemented — each bit of the encrypted message replaces the least significant bit of the R, G, and B channel of each pixel.
- **Capacity Check:** The system verifies the message fits within the image before encoding.
- **Format Handling:** Output saved as PNG to prevent JPEG compression from destroying the hidden bits.

Phase 3: Steganalysis & Testing (Evaluation)

- **PSNR measurement** computed and confirmed above 40 dB, validating imperceptibility.
- **Visual inspection** performed — original and stego images are indistinguishable to the human eye.
- **Encode/Decode verification** — the decoded message was confirmed to exactly match the original input.
- Note: Statistical steganalysis (e.g., chi-square test) and cover source mismatch testing are identified as areas for future work.

Phase 4: Ethical Review & Finalization (Deployment)

- The dual-use nature of the tool was assessed. The system is designed for legitimate privacy protection use cases such as securing confidential communications.
- Technical report and user documentation finalized.